🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Equations and Inequalities Problem Solving

Lesson 8-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can model and solve real-world problems using equations and inequalities.
Language Objective: I can explain my reasoning using the words model, equation, inequality, and reasonableness.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Model | Equation | Inequality | Reasonableness | Variable | Constraint

👆 *Tap any word to see its meaning and a picture.*

1. A drawing, equation, or diagram that represents a situation is a ____________.

2. A math sentence with an equal sign is an ____________.

3. A math sentence that compares values with <, >, ≤, or ≥ is an ____________.

4. Checking whether an answer makes sense for the problem is checking its ____________.

5. A letter that stands for the unknown amount in a problem is a ____________.

6. A limit that tells which values are allowed in a problem is a ____________________.

Write About the Math

How much more can the detective spend? How much does each flashlight cost? When do you use an equation versus an inequality?

¿Cuánto más puede gastar la detective? ¿Cuánto cuesta cada linterna? ¿Cuándo usas una ecuación y cuándo una desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Model** — *Modelar*
- ☐ **Equation** — *Ecuación*
- ☐ **Inequality** — *Desigualdad*
- ☐ **Reasonableness** — *Razonabilidad*
- ☐ **Variable** — *Variable*

Model: *An exact total points to an equation; a limit or range points to an inequality.* — Students use signal words ('equals/total' for an equation; 'at least/at most/no more than' for an inequality) to choose the correct model.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can spend at most $__ more because r ≤ 500 − ____ = ____. Each flashlight costs $__ because f = 60 ÷ ____ = ____.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Model
2. **Notes 2:** Equation
3. **Notes 3:** Inequality
4. **Notes 4:** Reasonableness
5. **Notes 5:** Variable
6. **Notes 6:** Constraint
7. **Watch for this mistake:** *A common mistake in Equations and Inequalities Problem Solving is matching the wrong model to a key phrase. For 'the getaway car held no more than 6 bags,' students often write the equation $b = 6$ instead of the inequality $b \leq 6$, because they see one number and assume it must be exact. The same mistake happens in reverse: for 'the reward split equally among 5 people gave each $40,' students sometimes write the inequality $5p \geq 40$ instead of the equation $5p = 40$, because 'split equally' sounds like it could mean a range. Before you submit, ask: does this phrase give one exact value (equation, =) or a limit/range (inequality, <, >, ≤, ≥)?*
8. **Try It 1:** A. $x \leq 20$ — *'At most 20' means 20 or fewer, so $x \leq 20$.*
9. **Try It 2:** A. $w = 12$ — *Divide both sides by 5: $w = 60 \div 5 = 12$ pounds. Check: $5 \times 12 = 60$ ✓*